

Jahangirnagar University
Institute of Information Technology

Course Code: ICT- 2203

Course Title: Operating Systems

Tutorial Question - 3

Total Marks: 10

Set-A

Date: 16/09/2025

Time: 30 mins.

1.	Define Multicore programming depending on the context of concurrency and parallelism.	3
2.	Briefly discuss about the types of Multicore programming.	3
3.	Discuss about Java Runnable Interface.	2
4.	Define many-to-one multithreaded models of thread.	2

Name :

roll:

ODD

1. If there is a network like 172.168.0.0/19, then determine the followings:

- I. How many subnets are possible?
- II. How many hosts are there?
- III. What is the subnet mask of the network?
- IV. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

In the 18th subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

Name :

roll:

EVEN

2. If there is a network like 172.168.0.0/20, then determine the followings:

- I. How many subnets are possible?
- II. How many hosts are there?
- III. What is the subnet mask of the network?
- IV. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

In the 18th subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

Tutorial

Jahangirnagar University
Department of IIT, Course # IIT 3109(Basic-Economics)
1st Tutorial Examination-2025, 51 batch
Marks: 20, Time: 55 minutes

1. a) What is economics? What are the basic differences between micro and macroeconomics? 5

b) What is the positive and normative statement of economics? Give example. 2

c) i) State the demand and supply law. ii) Given a demand equation $Q_d = 100 - 4P$; where Q_d is the quantity demand and P is the price and a supply equation $Q_s = 80 + P$; where Q_s is the quantity supplied and P is the price. Draw the demand and supply curve correctly and show the market equilibrium point and calculate the equilibrium price and quantity. 4+9=13

OR

c) i) Define market equilibrium. Suppose, A hypothetical market is in equilibrium position- If the market price will increase and will decrease from the equilibrium price, what will happen on the economy-explain (shortage and surplus) with figure.

ii) Use the demand and the supply diagram, what will happen in the economy? If the consumer's income will increase and raw material price will decrease in the economy. 9+4=13

Jahangirnagar University
2nd Tutorial Examination 2024
Department Of IIT
Course # IIT -3109 (Basic Economics)
Total Marks: 20, Time:45+45 minutes

4+6+10=20

01.a) Define GDP(Gross Domestic Products).What is GDP gap? Draw the business cycle and explain.

b) Which of the following transactions would be counted or would not be counted in GDP?

Explain your own answer:

- i. Bangladesh Govt. builds a new industrial plant.
- ii. A drug dealer sells \$ 5000 worth of illegal drugs.
- iii. Water and Air pollution decreases in Dhaka city.
- iv. Bangladesh earns a big amount of money from export sectors.

c) Below are some data on two goods produced in the economy.

Year	Price of rice(/kg) in Tk.	Quantity of rice (kg)	Price of cloth (/yard) in Tk.	Quantity of cloth (/yard)
2004	10	100	20	50
2005	10	200	20	100
2006	20	200	40	100

Compute nominal GDP, real GDP, GDP deflator, and the economic growth rate for each year, using 2004 as the base year.

4+6+10=20

02. a) What do you mean by short run and long run phenomenon? Explain fixed cost, variable cost and total cost.

b) Given the data, compute Total cost (TC), Marginal cost(MC), Average fixed cost(AFC), Average variable cost (AVC), Average total cost (ATC) and show those Cost curves graphically.

Combination.	Labor	Output	TFC	TVC
A	0	0	25	25 0
B	1	6	25	50
C	2	12	25	75
D	3	15	25	100
E	4	20	25	150
F	5	25	25	175
G	6	30	25	225

c) Given the following total revenue (TR) and total cost (TC) functions for a firm, maximize profit (π) for the firm as follows: (i)Find the average revenue (AR): average cost (AC) functions (ii) Set up the profit function(π). (iii) Find the critical value(s) where π is at a relative extremum and test second order condition and calculate the maximum profit.

$$TR = 20Q - 2Q^2 \quad TC = Q^2 + 4Q$$